

Def. Let  $K$  be a commutative Ring with identity.

A map  $D: M_{n \times n}(K) \rightarrow K$  is called  $n$ -linear if: for each  $1 \leq i \leq n$ ,

$$D \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ c\alpha_i + \alpha'_i \\ \vdots \\ \alpha_n \end{pmatrix} = c \cdot D \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_i \\ \vdots \\ \alpha_n \end{pmatrix} + D \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha'_i \\ \vdots \\ \alpha_n \end{pmatrix}$$

when  $c \in K$ ,  $\alpha_1, \dots, \alpha_n, \alpha'_i$  are arbitrary row vectors.

Def. A  $n$ -linear map  $D: M_{n \times n}(K) \rightarrow K$  is called Alternating if:

$$D(A) = 0 \text{ whenever two rows of } A \text{ are equal.}$$

Lemma. If  $n$ -linear map  $D: M_{n \times n}(K) \rightarrow K$  is Alternating,

then for  $A'$ , obtained from  $A$  by interchanging two rows of  $A$ ,  $D(A') = -D(A)$ .

Proof sketch. For odd  $n$

Def. Let  $K$  be a commutative ring with identity.

The unique map  $D: M_n(K) \rightarrow K$  satisfying

$n$ -linear, alternating,  $D(I) = 1$

is called the determinant function.

Example. The determinant of 2 by 2 matrix is determined by:  $\det A = A_{11}A_{22} - A_{12}A_{21}$ .

Given  $A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} = (A_{11}e_1 + A_{12}e_2, A_{21}e_1 + A_{22}e_2)$

Since  $D$  is  $n$ -linear,

$$\begin{aligned} D(A) &= D(A_{11}e_1 + A_{12}e_2, A_{21}e_1 + A_{22}e_2) \\ &= A_{11}D(e_1, A_{21}e_1 + A_{22}e_2) + A_{12}D(e_2, A_{21}e_1 + A_{22}e_2) \\ &= A_{11}A_{21}D(e_1, e_1) + A_{11}A_{22}D(e_1, e_2) + A_{12}A_{21}D(e_2, e_1) + A_{12}A_{22}D(e_2, e_2) \end{aligned}$$

Since  $D$  is alternating,  $D(e_1, e_1) = D(e_2, e_2) = 0$  and  $D(e_2, e_1) = -D(e_1, e_2) = -D(I) = -1$ .

So,

$$= A_{11}A_{22} - A_{12}A_{21} \quad \square$$

A matrix  $\text{adj } A = \begin{bmatrix} A_{22} & -A_{12} \\ -A_{21} & A_{11} \end{bmatrix}$  is called the classical adjoint of  $A$ .

Using the properties of  $\det$ ,  $n$ -linear, alternating,  $\det I = 1$ , deduce the following:

a).  $(\text{adj } A) \cdot A = A \cdot (\text{adj } A) = (\det A) \cdot I$ .

b).  $\det(\text{adj } A) = \det A$

c).  $\text{adj}(A^t) = (\text{adj } A)^t$ .

Proof. The first claim can be shown directly:  $(\text{adj } A) \cdot A = \begin{bmatrix} A_{22}A_{11} - A_{12}A_{21} & 0 \\ 0 & -A_{21}A_{12} + A_{11}A_{22} \end{bmatrix}$

Similarly b) and c).

Given a  $n \times n$  matrix  $A$  over a commutative ring with identity  $K$ , the  $i$ 'th row  $d_i$  is  $\sum_{j=1}^n A_{ij} e_j$  for  $1 \leq i \leq n$ .

$$d_i = \sum_{j=1}^n A_{ij} \cdot e_j.$$

Therefore, for  $n$ -linear, alternating function  $D$ ,

$$\begin{aligned} D(A) &= D(d_1, d_2, \dots, d_n) = \sum_{j_1=1}^n A_{1,j_1} \cdot D(e_{j_1}, d_2, \dots, d_n) \\ &= \sum_{j_1=1}^n A_{1,j_1} \cdot \sum_{j_2=1}^n A_{2,j_2} \cdot D(e_{j_1}, e_{j_2}, d_3, \dots, d_n) \\ &\quad \vdots \\ &= \sum_{1 \leq j_1, \dots, j_n \leq n} A_{1,j_1} \cdot A_{2,j_2} \cdot \dots \cdot A_{n,j_n} \cdot D(e_{j_1}, \dots, e_{j_n}) \end{aligned}$$

Since  $D$  is alternating, if there are two of same indices, then  $D(e_{j_1}, \dots, e_{j_n}) = 0$ .

So, we can consider  $(e_{j_1}, \dots, e_{j_n})$  as a permutation on  $n$  letters, moreover, if  $D(I) = 1$ , then  $D(e_{j_1}, \dots, e_{j_n}) = \text{sgn}(j_1, \dots, j_n)$ . In sum,

$$\det A = \sum_{\sigma \in S_n} \text{sgn } \sigma \cdot A_{1,\sigma(1)} \cdot \dots \cdot A_{n,\sigma(n)}.$$

✓ Invertibility of Matrix.

Lemma. Let  $K$  be a commutative Ring with identity, and let  $A, B \in M_{n \times n}(K)$ .

Then  $\det AB = \det A \cdot \det B$ .

Proof. Let  $B$  be a fixed  $n \times n$  matrix over  $K$ . Define  $D(A) = \det AB$ .

Denote  $\alpha_1, \dots, \alpha_n$  as  $i$ 'th rows of  $A$ . Then  $D(\alpha_1, \dots, \alpha_n) = \det(\alpha_1 B, \dots, \alpha_n B)$ .

Moreover,  $D$  is  $n$ -linear and alternating, since  $\det$  is. So,  $D(A) = \det A \cdot D(I) = \det A \cdot \det B$ .

Def. Given  $A \in M_{n \times n}(K)$ , define the Classical adjoint of  $A$  as a matrix

$$(\text{adj } A)_{ij} \stackrel{\text{def}}{=} (-1)^{i+j} \det A(j|i).$$

where  $A(j|i)$  is a  $(n-1)$  by  $(n-1)$  Matrix obtained by remove  $j$ 'th row and  $i$ 'th column.

Thm. Let  $K$  be a commutative Ring with identity.

Given  $A \in M_{n \times n}(K)$ ,

$$(\text{adj } A) \cdot A = (\det A) \cdot I.$$

Proof.

